

CSI4901 Capstone Project

Intelligent Feedback System for Enhancing Placement Experiences

21MID0149, JAISHWANTH A

Under the Supervision of

Kanagaraj K

Associate Professor SCOPE

School of Computer Science and Engineering (SCOPE)

M.Tech (Integrated)

in

Computer Science and Engineering

with specialisation in Data Science

School of Computer Science and Engineering (SCOPE)



Abstract

Accumulated data on student course feedback and placements/internships is valuable to evaluate the quality of our education, but analysing the course feedback provided by students through text is a manual process that takes a lot of time, is subjective, and does not scale well when students provide a large amount of feedback.

This project will develop an intelligent system which will provide real-time, automated sentiment analyses for both student feedback and placement related using classical machine learning techniques and hybrid approaches. The system will preprocess unstructured textual feedback using commonly accepted text normalisation techniques; then, the textual feedback features will be converted into a numeric format using TF-IDF feature extraction. The resulting feedback will be classified using a variety of machine learning classifiers, such as SVM, random forest, KNN, and logistic regression, in order to classify feedback into sentiment categories.

In order to improve the system's robustness and performance, ensemble learning techniques will also be used to compare various classification models based on standard metrics like accuracy, precision, recall, and F1 score. The feedback will also be grouped for analysis purposes based on the aspects of location, size of the company or job role.

Additionally, this project aims to support organisations while respecting their privacy obligations using ethical data methods. The system proposed will help educational institutions to analyse feedback from students systematically to identify possible improvements and support data-based decisions when improving the quality of education and placement success.